

CONDITIONAL STATEMENTS

- Prepared by Isaak Kamau

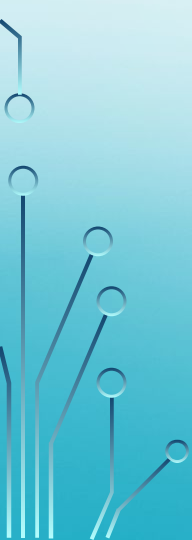
- Prepared by Isaak Kamau [linkedin.com/in/isaak-mwangi/](https://www.linkedin.com/in/isaak-mwangi/) | https://x.com/aidev_isaak

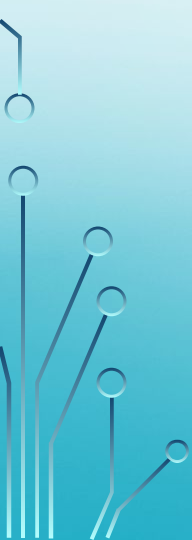


• In this session, we are shall be covering:

• Python Operators

• Python Conditional statements



- 
- 
- 
- 
- Conditional Statements Are part of Python CONTROL FLOW
 - Control flow describes the order in which your lines of code are run. This order is usually different than the sequence in which the lines of code appear! Execution can flow from one place in the code to another

INDENTATION

- Some other languages use braces to show where blocks of code begin and end. In Python, we use indentation to enclose blocks of code. For example, 'if' statements use indentation to tell Python what code is inside and outside of different clauses.
- In Python, indents conventionally come in multiples of four spaces. Be strict about following this convention, because changing the indentation can completely change the meaning of the code. If you are working on a team of Python programmers, it's important that everyone follows the same indentation convention!

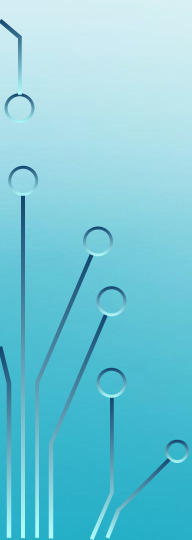
SPACES OR TABS?

The [Python Style Guide](#) recommends using 4 spaces to indent, rather than using a tab. Whichever you use, be aware that "Python 3 disallows mixing the use of tabs and spaces for indentation."

- Python supports the usual logical conditions from mathematics:
- Equals: $a == b$
- Not Equals: $a != b$
- Less than: $a < b$
- Less than or equal to: $a <= b$
- Greater than: $a > b$
- Greater than or equal to: $a >= b$
- These conditions can be used in several ways, most commonly in "if statements" and loops.



IF STATEMENT

- An “if” statement is a conditional statement that runs or skips code based on whether a condition is true or false.
- 
- 
- 



- Here's a simple example.

- `phone_balance = 20`

- `bank_balance = 1000`

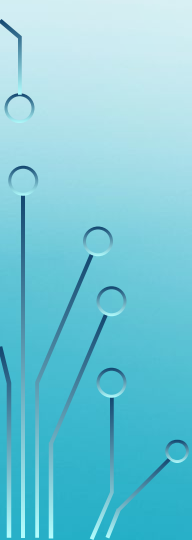
- `if phone_balance < 20:`

- `phone_balance += 50`

- `bank_balance -= 50`

-

- `print("Your phone_balance is: ", phone_balance)`





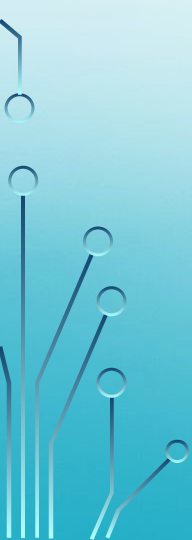
- `number = 145`

- `if number % 2 == 0: # == is called comparison operators`

- `print("Number " + str(number) + " is even.")`

- `else:`

- `print("Number " + str(number) + " is odd.")`



- To get inputs from the user we use “input()” function
- Example of how to get input from users
- `name = input('Please Enter your Name: ')`
- `print("Hello " + name + " and Welcome to Data Analysis Class")`

A PROGRAM FOR CHECKING WHETHER A NUMBER IS ODD OR EVEN

- `name = input('Please Enter your Name: ')`
- `number = input("Please Enter Your Number: ")`
- `number = int(number)`
- `if number % 2 == 0:`
 - `print("Hello " + name + "Your Number: " + str(number) + " is even.")`
- `else:`
 - `print("Hello " + name + " Your Number: " + str(number) + " is odd.")`

IF, ELIF, ELSE

- In addition to the if clause, there are two other optional clauses often used with an `if` statement.
- 1. `if`: An if statement must always start with an if clause, which contains the first condition that is checked. If this evaluates to True, Python runs the code indented in this if block and then skips to the rest of the code after the if statement.

ELIF

- 2. ``elif``: `elif` is short for "else if." An `elif` clause is used to check for an additional condition if the conditions in the previous clauses in the `if` statement evaluate to `False`. As you can see in the example, you can have multiple `elif` blocks to handle different situations.

ELSE

- 3. ``else``: Last is the else clause, which must come at the end of an if statement if used. This clause doesn't require a condition. The code in an else block is run if all conditions above that in the if statement evaluate to False.

NESTED IF

- You can have `if` statement inside `if` statement. This is called nested `if` Statements

EXAMPLE

- `x = 41`

```
if x > 10:  
    print("Above ten,")  
    if x > 20:  
        print("and also above 20!")  
    else:  
        print("but not above 20.")
```


- Simple Program:

- `season = input('What season are we in? ')`

- `if season == 'spring':`

- `print('plant the garden!')`

- `elif season == 'summer':`

- `print('water the garden!')`

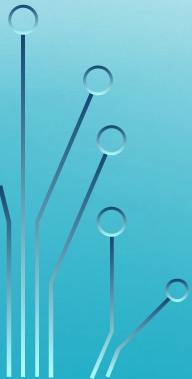
- `elif season == 'fall':`

- `print('harvest the garden!')`

- `elif season == 'winter':`

- `print('stay indoors!')`

- `else:`



- 
- 
- A simple project:
 - Let's combine all that we have learned and build a simple Calculator for addition, Subtraction, multiplication, and division “+- * /,”



- `num1 = int(input("num1: "))`

- `op = input("Operator: ")`

- `num2 = int(input("num2: "))`

- `if op == "+":`

- `print(num1 + num2)`

- `elif op == "-":`

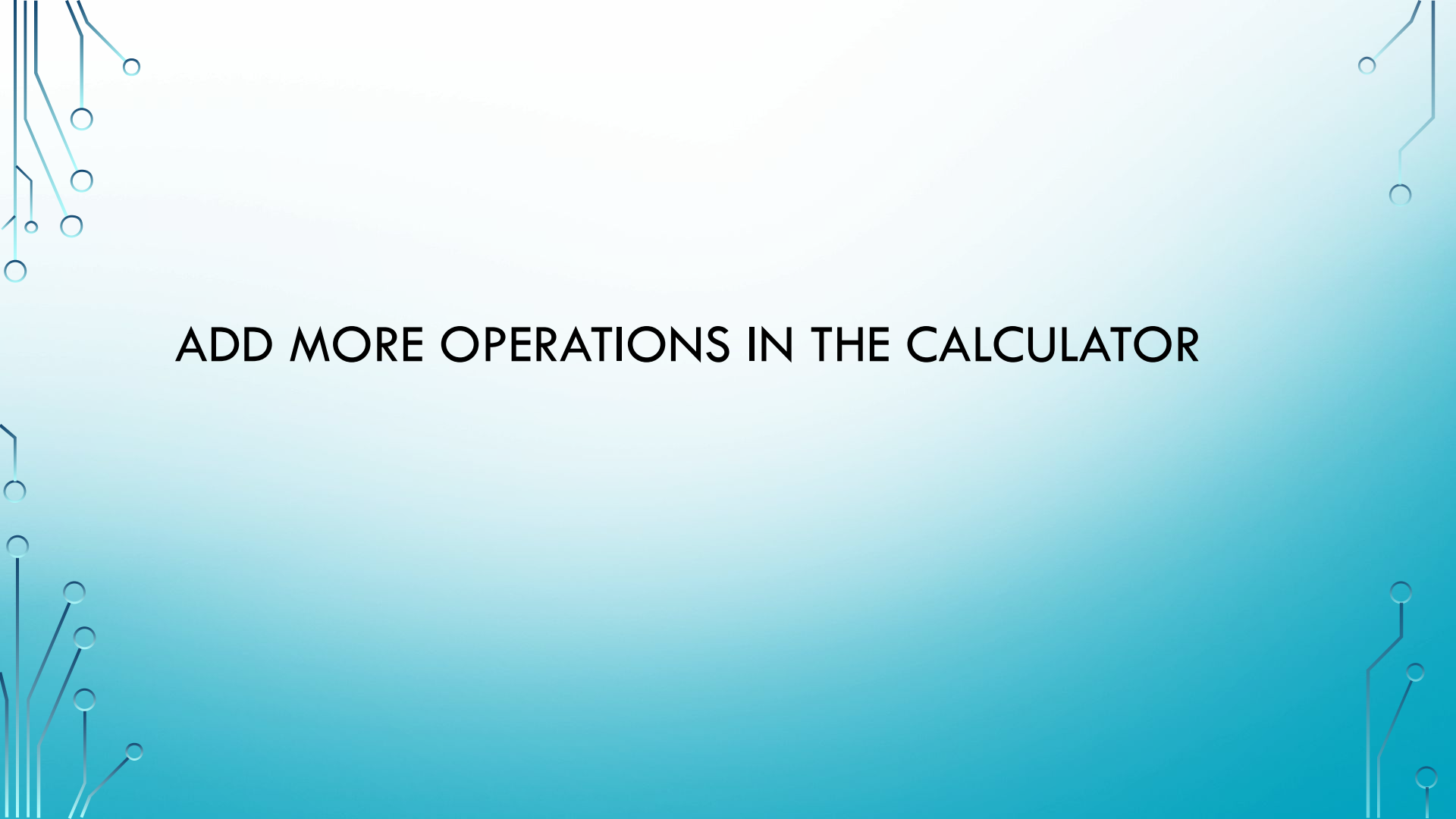
- `print(num1 - num2)`

- `elif op == "/":`

- `print(num1 / num2)`

- `elif op == "*":`

- `print(num1 * num2)`

The background is a light blue gradient. In the corners, there are decorative circuit-like patterns consisting of thin blue lines and small circles, resembling a stylized electronic board.

ADD MORE OPERATIONS IN THE CALCULATOR